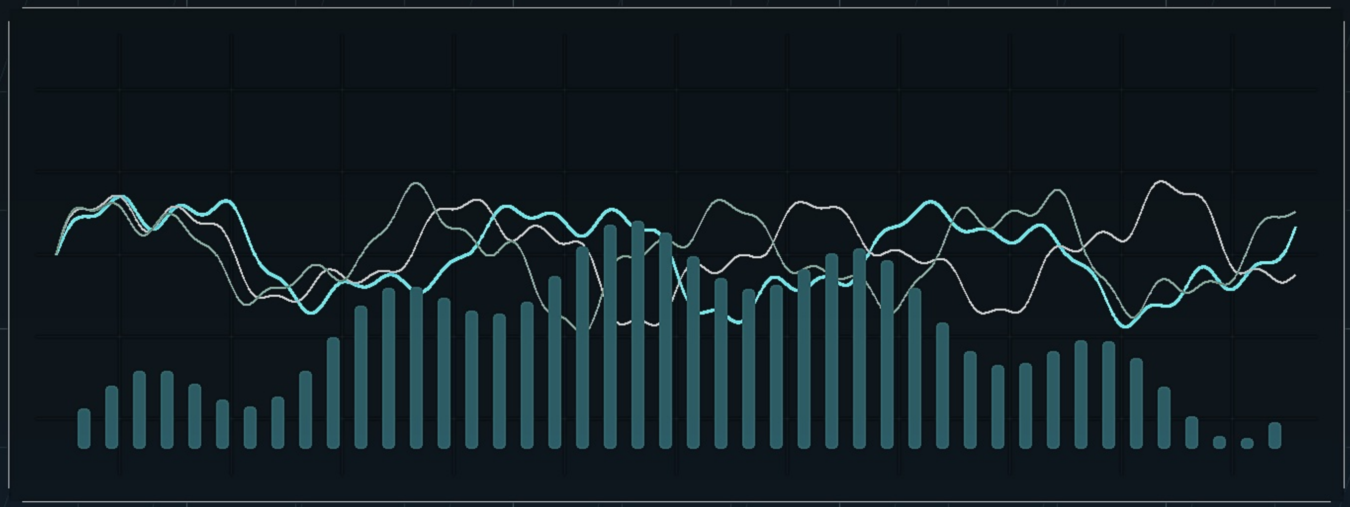
051-LAB ENGINEERING RECORDS

AXIOM-DSP

SESSION WORK LOG

JamesDSP / EEL2 Audio DSP



Current Baseline: v4.1.4.11

Stability First | Measurement Second | Experimentation Third

Repository workflow, qualification gates, Pi runbooks, and Codex orchestration record.

DOCUMENT CLASS: PUBLIC REPOSITORY LOG

NO AUDIO CAPTURES OR PRIVATE MATERIAL INCLUDED

Run 001 - AI Development Ecosystem Operating Framework

Date: 2026-06-03
Status: Completed
Scope: Documentation and workflow architecture only.

What Was Implemented

- Converted the local axiom-idea.html concept into repository-ready operating docs.
- Added the Codex, Pi, user, and external LLM role model.
- Added the system roadmap for Core, Reference, Immersive, Night, Studio Path, Labs, Knowledge, and Qualification.
- Added the ordered implementation plan for building the Axiom operating system.
- Added Labs policy, GitHub issue templates, Knowledge structure, Labs/review templates, and the system status dashboard.
- Updated the documentation index, agent instructions, contributing guide, harness documentation, and task backlog.

Why It Matters

- Axiom now has a controlled development structure around the DSP instead of loose one-off work.
- New ideas have clear intake paths before they can become DSP edits.
- Labs can explore without weakening the accepted Core baseline.
- External LLM feedback can be captured as advisory input and converted into testable work.
- A fresh Codex or Pi session can quickly find the accepted baseline, open investigations, and next work.

Validation

- git diff --check passed.
- GitHub issue-template YAML parsed successfully.
- Relative Markdown links resolved.
- Stale current-baseline scan found no current-facing .10 accepted-baseline references.
- No src/ or scripts/ files changed.

Next Recommended Work

- Review, commit, and publish the documentation and workflow foundation.
- Rerun the targeted .11 Sub Harmonics map when the JDSP route is available.
- Add concrete Knowledge or Labs examples only after the first real use case exists.

Run 002 - Session Work Log PDF Workflow

Date: 2026-06-03
Status: Completed
Scope: Documentation support tooling and generated PDF copies.

What Was Implemented

- Added docs/session-work-log.md as the editable source of truth for session work summaries.
- Added tools/generate_session_work_log_pdf.py to generate one PDF page per run section.
- Generated the repository PDF copy at docs/session-work-log.pdf.
- Generated the Windows Documents PDF copy at C:\Users\soloa\Documents\Axiom-DSP-Session-Work-Log.pdf.

Why It Matters

- The user now has one readable document for reviewing completed work outside the chat.
- The repo keeps the source log and a PDF copy, while Windows Documents keeps an easy-access mirror.
- Future sessions can append a new ## Run section and regenerate both PDF copies with one command.

Validation

- PDF generation uses only the Python standard library.
- Both destination PDFs are written by the same generator from the same source.
- Each run section starts on its own PDF page.

Next Recommended Work

- After each completed work section, append a new run entry to this source file.
- Regenerate the PDFs before final summaries or before publishing workflow changes.

Run 003 - Profile Matrix, Listening Protocol, And Release Gates

Date: 2026-06-03
Status: Completed
Scope: System architecture and workflow documentation only.

What Was Implemented

- Added docs/profile-matrix.md to define each Axiom lane, its current status, operating authority, allowed changes, and required promotion tests.
- Added docs/listening-protocol.md to define when to listen, how to set up comparisons, what material classes to use, and how to record accept/reject decisions.
- Added docs/release-gates.md to define documentation, tooling, Labs, candidate, listening, and accepted-baseline promotion gates.
- Updated docs/README.md, AGENTS.md, CONTRIBUTING.md, the operating-system implementation plan, and the task backlog so these gates are discoverable.

Why It Matters

- Future ideas now have a clear destination before code changes start.
- Listening work is separated into setup, material selection, comparison method, observation fields, and decision rules.
- The accepted baseline has explicit promotion gates, which protects Axiom Core from broad or under-tested changes.
- Profile concepts like Reference, Immersive, Night, and Studio Path can be discussed without accidentally becoming official products.

Validation

- git diff --check passed.
- Relative Markdown links resolved.
- GitHub issue-template YAML still parsed successfully.
- Stale current-baseline scan remained clean.
- No src/ or scripts/ files changed.

Next Recommended Work

- Add Pi runbooks for repeatable terminal execution.
- Then review and publish the full workflow foundation as a docs-focused PR.

Run 004 - Pi Runbooks

Date: 2026-06-03
Status: Completed
Scope: Terminal workflow architecture only.

What Was Implemented

- Added docs/pi-runbooks.md with repeatable missions for Codex/Pi sessions.
- Covered baseline audit, candidate readiness, investigation without candidate, Labs experiments, candidate qualification, listening acceptance, advisory review triage, repository housekeeping, and publication/merge.
- Updated docs/README.md, AGENTS.md, docs/system-status.md, the operating-system implementation plan, and the task backlog so the runbooks are discoverable.

Why It Matters

- A Pi session can now be opened with a narrow mission instead of vague context.
- Each runbook defines inputs, commands, outputs, stop conditions, and decisions.
- Real-host JDSP work is protected from overlapping measurements.
- Candidate creation, Labs work, advisory feedback, and publication now have repeatable terminal paths.

Validation

- git diff --check passed.
- Relative Markdown links resolved.
- GitHub issue-template YAML still parsed successfully.
- Stale current-baseline scan remained clean.
- No src/ or scripts/ files changed.

Next Recommended Work

- Review the full documentation/workflow batch.
- Commit and publish it as a docs-focused PR after approval.

Run 005 - Review, Commit, And Publish Workflow Foundation

Date: 2026-06-03
Status: Completed
Scope: Repository publication workflow for the documentation and architecture batch.

What Was Implemented

- Reviewed the full documentation/workflow scope before publishing.
- Re-ran validation for Markdown whitespace, issue-template YAML, relative links, PDF generation, stale baseline references, and accidental DSP/script changes.
- Created branch codex/docs-axiom-operating-system.
- Committed the workflow foundation with message docs(workflow): add Axiom operating system framework.
- Pushed the branch to GitHub.
- Opened PR #10: [codex] Add Axiom operating system workflow docs.

Why It Matters

- The Axiom operating-system foundation is now reviewable in GitHub instead of existing only as local edits.
- The PR keeps this broad documentation/workflow batch separate from main until reviewed and merge-approved.
- The validation record gives future agents confidence that no accepted DSP files were touched.

Validation

- git diff --check passed.
- Issue-template YAML parsed successfully.
- Relative Markdown links resolved.
- The session PDF regenerated and both PDF copies matched by SHA-256.
- pdfinfo confirmed the session PDF was valid.
- The stale .10 current-baseline scan was clean.
- git diff -- src scripts | wc -c returned 0.
- GitHub reported PR #10 as clean and mergeable.

Next Recommended Work

- Review PR #10.
- Merge only after explicit approval.
- After merge, return to the roadmap and choose the next workflow refinement or targeted .11 measurement.

Run 006 - PR Review Validation Cleanup

Date: 2026-06-03
Status: Completed
Scope: Pull request validation cleanup for the documentation/workflow batch.

What Was Implemented

- Reviewed PR #10 from the committed diff rather than only the local worktree.
- Fixed committed whitespace issues in issue templates, documentation files, and repository templates.
- Updated the session-log PDF generator so generated PDF xref entries no longer create Git whitespace warnings.
- Regenerated the repository session PDF and the Windows Documents mirror.

Why It Matters

- The PR now meets the same whitespace standard that will be enforced on merge.
- The PDF generator fix prevents this exact review failure from returning in future session-log updates.
- The cleanup keeps the workflow foundation publication-ready without changing any EEL DSP scripts.

Validation

- git diff --check main..HEAD passed after cleanup.
- Issue-template YAML parsed successfully.
- Relative Markdown links resolved.
- The session PDF regenerated and both PDF copies matched by SHA-256.
- pdfinfo confirmed the session PDF was valid.
- No src/ or scripts/ files changed.

Next Recommended Work

- Push the amended PR branch.
- Merge PR #10 only after explicit approval.

Run 007 - Post-Merge Roadmap Completion Reconciliation

Date: 2026-06-03
Status: Completed
Scope: Repository workflow roadmap status after PR #10 merge.

What Was Implemented

- Confirmed PR #10 merged into main as commit c498688.
- Confirmed the local main branch fast-forwarded to origin/main.
- Reconciled the completed workflow-foundation items that are now live in the repository:
- AI development ecosystem docs;
- Codex operating guide;
- Pi runbooks;
- Labs policy;
- Knowledge Base structure;
- profile matrix;
- release gates;
- listening protocol;
- GitHub issue templates;
- task backlog;
- session work log PDF workflow.

Why It Matters

- The PDF now has a single page that proves which roadmap foundation items are complete.
- Future Codex, Pi, and external-review sessions can distinguish completed workflow infrastructure from larger product-profile work that is only defined.
- The repository is positioned to move from workflow setup back into measured DSP investigation without losing the architecture record.

Validation

- PR #10 is merged on GitHub.
- Local main is synced with origin/main.
- Post-merge whitespace, issue-template YAML, and PDF validity checks passed.
- No src/ or scripts/ files changed in the workflow-foundation merge.

Next Recommended Work

- Refresh docs/system-status.md and docs/task-backlog.md so they no longer describe merged workflow items as pending.
- Then continue with the targeted .11 Sub Harmonics / limiter-pressure investigation or the next roadmap housekeeping task.

Run 008 - Post-Merge Reconciliation Cleanup

Date: 2026-06-03
Status: Completed
Scope: Documentation status reconciliation after the operating-system foundation merge.

What Was Implemented

- Updated docs/axiom-operating-system-implementation-plan.md with a completion summary for all eight implementation-plan sections.
- Updated docs/task-backlog.md with explicit status values for all 17 workflow-foundation tasks.
- Updated docs/system-status.md so the operating-system foundation is marked as merged instead of drafted.
- Preserved the distinction between completed workflow infrastructure and larger product lanes that are defined but not yet built.

Why It Matters

- A fresh agent can now tell that the implementation-plan phase is functionally complete.
- The backlog no longer points at already-merged publication work as the next priority.
- The status dashboard now sends the project toward full-state assessment, targeted .11 investigation, or Knowledge/Labs examples instead of stale PR work.

Validation

- Stale "drafted" and "review/publish" language was removed from active status docs.
- Historical session-log entries were left intact because they describe earlier run states.
- git diff --check passed after the reconciliation edits.
- The session PDF regenerated successfully in the repo and Windows Documents mirror.
- No EEL DSP scripts changed.

Next Recommended Work

- Review and commit the reconciliation cleanup.
- Then assess the full Axiom state before choosing the next DSP or workflow task.

Run 009 - Session Work Log Cover Upgrade

Date: 2026-06-04
Status: Completed
Scope: Documentation presentation upgrade only.

What Was Implemented

- Added deterministic technical-manual cover artwork for the session work-log PDF.
- Added tools/generate_session_cover_art.py so the cover asset can be regenerated locally.
- Updated tools/generate_session_work_log_pdf.py to embed the cover image as page 1 when the asset exists.
- Kept searchable cover text in the PDF without adding unsupported audio-quality claims.
- Regenerated the repository PDF and Windows Documents mirror.

Why It Matters

- The session work log now reads more like an official Axiom engineering record.
- The cover is reproducible and does not depend on private images, source audio, or external design files.
- Normal PDF refreshes still work from the repo and keep both PDF copies synchronized.

Validation

- Cover artwork generated successfully.
- PDF regenerated successfully with the cover as page 1.
- Repo and Windows PDF copies matched by SHA-256.
- PDF text extraction found the cover title.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the full local documentation batch.
- Then assess the current Axiom system state before starting the next major agentic-system functionality set.

Run 010 - Agentic Engineering Blueprint Started

Date: 2026-06-04
Status: Completed
Scope: New living blueprint for the future Axiom agentic engineering system.

What Was Implemented

- Added docs/axiom-agentic-engineering-blueprint.md as the source document for the agentic-system architecture.
- Added tools/generate_agentic_blueprint_pdf.py to generate a matching PDF.
- Updated docs/README.md so the blueprint is discoverable from the documentation index.
- Generated the repository PDF at docs/axiom-agentic-engineering-blueprint.pdf.
- Generated the Windows Documents mirror at C:\Users\soloa\Documents\Axiom-DSP-Agentic-Engineering-Blueprint.pdf.
- Captured the current gap: Axiom has Pi roles and commands, but not native Codex Axiom agents, subagents, or commands yet.

Why It Matters

- The agentic-system idea now has its own durable architecture document instead of living only in chat.
- The blueprint separates the Human Authority, Codex Orchestration, Pi Execution, and Knowledge/Advisory layers.
- The next major functionality set can be planned against a written target: agent registry, Codex skill pack, knowledge retrieval, role reviews, approval gates, and handoff protocol.

Validation

- Blueprint PDF generated successfully in the repo and Windows Documents mirror.
- PDF text extraction found the blueprint title.
- git diff --check passed after adding the blueprint files.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the blueprint and decide the first implementation phase for the Axiom Agentic Engineering Layer.

Run 011 - Axiom Agentic Engineering Layer v1

Date: 2026-06-04
Status: Completed
Scope: Repo-tracked native Codex orchestration layer and safe helper tooling.

What Was Implemented

- Added repo-tracked Axiom Codex skill source under tools/codex-skills/axiom-engineering/.
- Added skill references for role registry, handoff protocol, approval matrix, Knowledge policy, and helper CLI usage.
- Added tools/install_axiom_codex_skill.py with --dry-run and explicit --install modes.
- Added tools/axiom-codex/axiom_codex.py with status-summary, ready-check, agent-review, and knowledge-query commands.
- Added the local-only Knowledge source-index schema and a repo-safe source-note template.
- Updated the agentic blueprint, documentation index, tool inventory, Knowledge docs, and system status.

Why It Matters

- Axiom now has a native Codex-side orchestration layer in source form without replacing the Pi execution harness.
- Codex can summarize state, frame multi-role reviews, and search Knowledge safely while real-JDSP and candidate workflows remain delegated to Pi.
- The skill can be previewed before installation, keeping local Codex changes explicit and reversible.
- Knowledge integration now supports both public repo-safe notes and a private local source index.

Validation

- Axiom Codex helper CLI smoke tests passed.
- Skill install dry run listed the expected files without modifying ~/.codex.
- Python helper scripts compiled.
- git diff --check passed.
- Markdown relative links resolved.
- Blueprint and session PDFs regenerated successfully in repo and Windows Documents mirrors.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the local v1 agentic layer.
- Install the Codex skill only after explicit approval.
- Then decide whether to deepen Knowledge ingestion or start a measured Axiom DSP assessment.

Run 012 - Full Axiom System Assessment

Date: 2026-06-04
Status: Completed
Scope: Current-state assessment across Core DSP, Qualification, Agentic tooling, Knowledge, and repository housekeeping.

What Was Implemented

- Added docs/axiom-full-state-assessment-2026-06-04.md.
- Linked the assessment from docs/README.md.
- Updated docs/system-status.md with the latest assessment summary and next actions.
- Updated docs/task-backlog.md so the full-state assessment is marked as the current checkpoint instead of pending work.
- Captured a validation snapshot for the accepted v4.1.4.11 Core baseline, Pi harness, material corpus, candidate readiness, and Codex helper layer.

Why It Matters

- Axiom now has a single current-state assessment that says what is accepted, what is strong, what is still weak, and what should not happen next.
- The assessment keeps .11 as the accepted baseline while making clear that candidate readiness does not justify creating .12 without a scoped hypothesis.
- The document separates the local agentic layer's real current state from the future target: source-ready and validated, but not installed as a native Codex runtime yet.
- The roadmap can now move from broad operating-system buildout into focused follow-through: close the .11 Sub Harmonics boundary, optionally install the Codex skill, and populate Axiom Knowledge.

Validation

- scripts/validate_axiom_static.sh src/axiom_binaural_dsp_v4.1.4.11.eel passed.
- python3 -m unittest discover -s tests -p 'test_*.py' passed with 159 tests.
- npm test in tools/axiom-team passed with 22 tests.
- node tools/axiom-team/bin/axiom-team.mjs doctor passed.
- node tools/axiom-team/bin/axiom-team.mjs corpus-status --strict-metadata passed with 14 tracks and required metadata coverage.
- python3 scripts/evaluate_axiom_candidate_readiness.py reported READY.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- git diff --check passed.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the local docs/tooling batch.
- Commit and publish the batch when explicitly approved.
- Install the Axiom Codex skill only after explicit approval.
- Run the targeted .11 Sub Harmonics map when the JDSP route is available.
- Add the first lawful Axiom Knowledge source notes.

Run 013 - Local Docs And Agentic Tooling Pre-Commit Audit

Date: 2026-06-04
Status: Completed
Scope: Pre-commit review cleanup for the local documentation and agentic-tooling batch.

What Was Implemented

- Audited the current local diff and untracked files before any commit or publication.
- Confirmed the active diff remains documentation and tooling only.
- Updated the Axiom Codex helper agent-review scaffold so it no longer emits weak TBD placeholders.
- Updated docs/system-status.md to describe the agentic blueprint, skill source, and helper CLI as source-ready instead of merely drafted.
- Removed generated Python cache directories created by validation runs.

Why It Matters

- The local batch is closer to review-ready without changing the accepted DSP baseline.
- The agent-review helper now makes clear that role findings must be completed before the scaffold can be used as evidence.
- The system-status wording better distinguishes the real current state from the future target: source-ready local tooling exists, but the Codex skill is not installed yet.
- The repo tree is cleaner for review because generated runtime cache files were removed.

Validation

- Axiom Codex skill validation passed with quick_validate.py.
- python3 tools/axiom-codex/axiom_codex.py agent-review --topic "local docs tooling pre-commit audit" produced the updated scaffold.
- git diff --check passed.
- Python helper scripts compiled.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the local docs/tooling batch as the candidate commit scope.
- Commit and publish only after explicit user approval.
- Keep DSP work paused until the documentation/tooling batch is safely recorded or the user explicitly redirects.

Run 014 - Agentic Foundation Commit And Draft PR

Date: 2026-06-04
Status: Completed
Scope: Local commit and GitHub publication for the docs/tooling agentic-foundation batch.

What Was Implemented

- Created branch codex/agentic-engineering-foundation.
- Staged only the intended documentation, generated PDF, generated cover-art, Knowledge template/schema, Axiom Codex helper, Codex skill source, and PDF-generator tooling files.
- Committed the batch as cef0fee with message Add agentic engineering foundation.
- Pushed the branch to origin/codex/agentic-engineering-foundation.
- Opened draft PR #11: https://github.com/051-lab/axiom-binaural-dsp/pull/11.

Why It Matters

- The agentic engineering foundation is now available on GitHub for review from other devices and sessions.
- Publication happened through a branch and draft PR instead of direct main mutation.
- Merge remains a separate explicit approval gate.
- The accepted Axiom Core DSP baseline remains unchanged.

Validation

- gh auth status confirmed GitHub authentication for 051-lab.
- git diff --cached --check passed.
- git diff --check passed.
- Python helper scripts compiled.
- Axiom Codex skill validation passed.
- Markdown relative-link validation passed.
- npm test in tools/axiom-team passed with 22 tests.
- python3 -m unittest discover -s tests -p 'test_*.py' passed with 159 tests.
- git diff -- src scripts | wc -c returned 0.
- PR #11 is open as a draft and GitHub reports it as mergeable.

Next Recommended Work

- Review PR #11.
- Merge only after explicit user approval.
- After merge, install the local Axiom Codex skill only if the user approves that runtime change.
- Then resume targeted .11 measurement or Axiom Knowledge population.

Run 015 - Axiom Codex Skill Local Runtime Activation

Date: 2026-06-07
Status: Completed
Scope: Local Codex runtime activation and verification only.

What Was Implemented

- Installed the repo-tracked Axiom Codex skill into the local Codex runtime at ~/.codex/skills/axiom-engineering.
- Confirmed the installed local skill copy contains SKILL.md, agents/openai.yaml, and the five reference files.
- Verified a fresh Codex prompt context advertises $axiom-engineering as an available skill.
- Re-ran the Axiom Codex helper status and readiness checks.
- Re-ran the Pi harness doctor check against the accepted v4.1.4.11 baseline.

Why It Matters

- Codex can now load the Axiom-specific orchestration rules directly from the local runtime instead of relying only on repo docs.
- Future Axiom sessions can start with $axiom-engineering to apply the accepted baseline, handoff, approval, Knowledge, and safety policies.
- The install keeps Pi as the controlled real-JDSP and candidate execution layer.
- This activation does not add native Axiom slash commands, custom subagents, or any sound-changing behavior.

Validation

- python3 tools/install_axiom_codex_skill.py --install completed successfully.
- codex debug prompt-input 'Use $axiom-engineering to inspect Axiom state.' listed axiom-engineering.
- python3 tools/axiom-codex/axiom_codex.py status-summary reported accepted baseline v4.1.4.11, no active candidate, and zero DSP/script diff files.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- node tools/axiom-team/bin/axiom-team.mjs doctor passed and verified the accepted baseline hash.
- No EEL DSP scripts changed.

Next Recommended Work

- Use $axiom-engineering in fresh Codex sessions for Axiom work.
- Decide whether the next Codex-layer iteration should add command surfaces, role-specific subagents, or deeper Knowledge lookup.
- Resume the targeted .11 Sub Harmonics / limiter-pressure investigation when the JDSP route is available.

Run 016 - Codex Command Surface, Guardrails, Profiles, And Skill Evals

Date: 2026-06-07
Status: Completed
Scope: Codex-layer hardening for Axiom orchestration only.

What Was Implemented

- Added a repo-tracked Axiom Codex command registry at tools/axiom-codex/command_surface.json.
- Extended tools/axiom-codex/axiom_codex.py with command-surface, agent-profiles, guard-check, and skill-eval.
- Added Codex-specific role profile specs under tools/axiom-codex/agent_profiles/.
- Added deterministic skill behavior fixtures at tools/axiom-codex/skill_eval_cases.json.
- Added focused unit coverage in tests/test_axiom_codex_helper.py.
- Updated the Axiom skill references, tool inventory, system dashboard, and backlog so AX-TASK-018 through AX-TASK-021 are initial implementations instead of proposed gaps.
- Reinstalled the local Axiom Codex skill copy from the updated repo source.

Why It Matters

- Axiom now has a concrete command surface for repeated Codex workflows without relying on undocumented native slash-command behavior.
- Role-specific Codex profiles make coordination, DSP architecture, EEL safety, measurement, qualification, release, tooling, research, safety, and implementation responsibilities explicit.
- The guard check catches known unsafe publication scope before it can become a commit or PR problem.
- The skill eval fixtures give future sessions a deterministic way to confirm that the skill still encodes the expected Axiom behaviors.

Validation

- python3 -m unittest discover -s tests -p 'test_*.py' passed with 165 tests.
- python3 tools/axiom-codex/axiom_codex.py command-surface listed 10 commands.
- python3 tools/axiom-codex/axiom_codex.py agent-profiles listed 10 profiles.
- python3 tools/axiom-codex/axiom_codex.py guard-check passed with no known unsafe changed paths.
- python3 tools/axiom-codex/axiom_codex.py skill-eval passed 7 behavior fixtures.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- python3 tools/install_axiom_codex_skill.py --install --force synced the local runtime skill copy.
- codex debug prompt-input 'Use $axiom-engineering to run an Axiom guard check.' listed the updated axiom-engineering skill in fresh prompt context.
- No EEL DSP scripts changed.

Next Recommended Work

- Use command-surface, agent-profiles, guard-check, and skill-eval in future Axiom Codex sessions.
- Add native wrappers only when Codex exposes a supported slash-command or subagent mechanism that can preserve these boundaries.
- Resume the targeted .11 Sub Harmonics / limiter-pressure investigation when the JDSP route is available.

Run 017 - Read-Only Pi Handoff Helper

Date: 2026-06-07
Status: Completed
Scope: Codex-to-Pi handoff preparation only.

What Was Implemented

- Added python3 tools/axiom-codex/axiom_codex.py pi-handoff.
- Made the default handoff target the current .11 Sub Harmonics investigation with map-sub-gain at +10 dB and +12 dB.
- Updated the command registry and Axiom skill helper reference so pi-handoff appears as a Codex helper.
- Added targeted tests for default and custom handoff command generation.

Why It Matters

- Codex can now prepare the exact Pi command, preconditions, artifact policy, and approval boundary without running real JDSP.
- The next measurement step is easier to review before execution.
- Candidate creation, publication, merge, and accepted-baseline changes remain separate approval gates.

Validation

- python3 -m unittest tests.test_axiom_codex_helper passed with 9 tests.
- python3 tools/axiom-codex/axiom_codex.py pi-handoff printed the draft handoff and did not execute the Pi command.
- No EEL DSP scripts changed.

Next Recommended Work

- Review the generated handoff brief.
- Run the Pi command only when the JDSP route is available and the user explicitly approves real-host measurement.

Run 018 - Spatial Hearing Knowledge Source Note

Date: 2026-06-07
Status: Completed
Scope: Repo-safe Knowledge note creation from a local-only PDF source.

What Was Implemented

- Copied the user-provided Spatial Hearing - Revised Edition.pdf into the local-only Axiom Knowledge source store outside the repo.
- Added the spatial-hearing-revised-edition entry to the local-only source index.
- Created docs/knowledge/spatial-hearing-revised-edition.md with citation metadata, original Axiom-relevance notes, follow-up questions, and evidence boundaries.
- Updated the system dashboard and backlog so Axiom Knowledge reflects the first source-backed note.

Why It Matters

- Axiom now has a lawful Knowledge starting point for spatial hearing, localization vocabulary, headphone image stability, and crossfeed-boundary thinking.
- The repo note avoids copying book text and does not expose private local paths.
- The source can inform test design, but it does not prove any Axiom behavior.

Validation

- python3 tools/axiom-codex/axiom_codex.py knowledge-query "spatial hearing localization binaural" found the local source index entry.
- No EEL DSP scripts changed.

Next Recommended Work

- Use the Spatial Hearing note to refine future listening-record vocabulary.
- Keep adding Knowledge notes only as short original summaries with clear Axiom test-design questions.

Run 019 - Axiom Knowledge PDF Candidate Intake

Date: 2026-06-07
Status: Completed
Scope: Local-only PDF intake and repo-safe seed bibliography notes.

What Was Implemented

- Copied five user-provided PDF candidates from the desktop Knowledge folder into the local-only Axiom Knowledge source store outside the repo.
- Registered the five sources in the local-only source index: accurate-sound-reproduction-using-dsp, dafx-digital-audio-effects, designing-audio-effect-plug-ins-in-cpp, principles-and-applications-of-spatial-hearing, and the-audio-programming-book.
- Added repo-safe Knowledge notes for each source under docs/knowledge/.
- Updated docs/knowledge/README.md, docs/system-status.md, and docs/task-backlog.md to reflect the six-source Knowledge seed set.

Why It Matters

- Axiom Knowledge now has seed references for spatial hearing, digital audio effects, real-time implementation patterns, reproduction accuracy, and general audio-programming background.
- The PDFs remain local-only; the repo contains citation metadata, original summaries, Axiom relevance, and test-design questions.
- These notes provide vocabulary and review context without claiming that any source proves Axiom behavior.

Validation

- python3 tools/axiom-codex/axiom_codex.py knowledge-query "DAFX spatial hearing audio programming DSP" found the new local source-index entries.
- No EEL DSP scripts changed.

Next Recommended Work

- Use the seed notes to create short concept notes only when a specific Axiom test, review, or listening question needs them.
- Keep raw PDFs and private local paths out of git.

Run 020 - Knowledge Source Audit Helper

Date: 2026-06-08
Status: Completed
Scope: Codex-side Knowledge source integrity tooling.

What Was Implemented

- Added python3 tools/axiom-codex/axiom_codex.py knowledge-sources.
- The helper audits the local-only source index, required source fields, local file existence, duplicate IDs, source status/type values, and matching repo-safe notes.
- Private localPath values stay hidden unless --show-private-paths is explicitly used.
- Added targeted unit tests for missing local files and private-path hiding.
- Updated the command registry, helper reference, and tool inventory.

Why It Matters

- Axiom can now verify that local PDF sources and repo-safe Knowledge notes still line up after future source additions or machine moves.
- Knowledge source health can be checked without committing PDFs or exposing private paths.
- This makes the local-source plus repo-note workflow repeatable for future research intake.

Validation

- python3 -m unittest tests.test_axiom_codex_helper passed with 11 tests.
- python3 tools/axiom-codex/axiom_codex.py knowledge-sources reported 6 checked sources with local files and repo notes.
- No EEL DSP scripts changed.

Next Recommended Work

- Use knowledge-sources after adding or moving local PDFs.
- Convert seed bibliography entries into short concept notes only when a specific Axiom engineering question needs them.

Run 021 - Full-System Readiness Review

Date: 2026-06-08
Status: Completed
Scope: Evidence-local full-system readiness review across Core DSP, Qualification, Pi/JDSP workflow, Codex tooling, Knowledge, documentation, and release posture.

What Was Implemented

- Added docs/axiom-full-system-review-2026-06-08.md.
- Updated docs/README.md and docs/system-status.md so the June 8 review is the current assessment checkpoint.
- Added AX-TASK-022 through AX-TASK-026 to docs/task-backlog.md.
- Re-ran the local evidence snapshot without running real JDSP or creating a candidate.

Why It Matters

- Axiom now has a current, findings-first review that names strengths and shortcomings honestly.
- The review keeps .11 as accepted, treats candidate readiness as a gate rather than justification, and identifies the open .11 Sub Harmonics follow-up as the main Core evidence gap.
- The next improvement set is visible in the backlog before any .12 discussion.

Validation

- python3 -m unittest discover -s tests -p 'test_*.py' passed with 170 tests.
- python3 tools/axiom-codex/axiom_codex.py ready-check passed.
- python3 tools/axiom-codex/axiom_codex.py guard-check passed.
- python3 tools/axiom-codex/axiom_codex.py skill-eval passed.
- python3 tools/axiom-codex/axiom_codex.py knowledge-sources passed with 6 checked sources.
- node tools/axiom-team/bin/axiom-team.mjs doctor passed.
- node tools/axiom-team/bin/axiom-team.mjs corpus-status --strict-metadata passed with 14 tracks.
- python3 scripts/evaluate_axiom_candidate_readiness.py reported READY.
- PR #12 remains an open draft with clean merge state.
- No EEL DSP scripts changed.

Next Recommended Work

- Review and merge PR #12 only after explicit approval.
- Run the targeted .11 Sub Harmonics follow-up only when the JDSP route is available and real-host measurement is approved.
- Add structured spatial listening vocabulary before any new sound-changing candidate.

Run 022 - Sub Harmonics Follow-Up Gate And Listening Vocabulary

Date: 2026-06-08
Status: Partially completed
Scope: Targeted .11 Sub Harmonics follow-up execution path, command correction, and structured spatial listening vocabulary.

What Was Implemented

- Attempted the targeted .11 Sub Harmonics map through the Node harness.
- Identified that targeted maps must include the accepted +4 dB default comparison point, not only +10 dB and +12 dB.
- Updated the Codex pi-handoff helper and Node harness wrapper so targeted map commands include +4 dB when needed.
- Corrected the dashboard, harness docs, Pi runbook, and previous assessment command examples.
- Added required listening-record fields for lateral spread, localization blur, depth impression, bass-image coupling, and route context.
- Updated the Android validation package template and validator tests to match the expanded listening-record schema.

Why It Matters

- The .11 Sub Harmonics boundary remains open for the right reason: route availability, not a proven audio defect.
- Future targeted map commands should no longer fail argument validation because the default control point is missing.
- Future listening records can describe spatial effects with enough precision to support candidate decisions instead of relying on vague width comments.

Validation

- The corrected targeted map reached the harness but failed before measurement because the JDSP capture route was unavailable.
- No .12 candidate was created.
- No EEL DSP scripts changed.

Next Recommended Work

- Restore the JDSP capture route, then rerun the targeted .11 Sub Harmonics map at +4, +10, and +12 dB.
- Keep AX-TASK-022 route-blocked until that real-host measurement completes.
- Continue with PR #12 review/merge only after explicit approval.

Run 023 - Completed ‘.11‘ Sub Harmonics Follow-Up

Date: 2026-06-08
Status: Completed measurement; interpretation open
Scope: Restore JDSP capture route, rerun the corrected targeted .11 Sub Harmonics map, and record repo-safe evidence.

What Was Implemented

- Restored the private JDSP Pulse route with the configured jdsp-audio-reset helper.
- Verified WinSink, JamesDSP, and JamesDSP.monitor on the private unix:/tmp/jdsp-win/native server.
- Reran the corrected targeted map at +4, +10, and +12 dB using the dense/flawed/bass-oriented material filter.
- Added docs/sub-harmonics-follow-up-v4.1.4.11.md as the repo-safe summary of the local report.
- Updated docs/system-status.md, docs/task-backlog.md, and docs/README.md to point future sessions at the completed follow-up evidence.

Why It Matters

- AX-TASK-022 is no longer route-blocked; it has a completed real-JDSP follow-up map.
- The result did not show normal-material clipping through +12 dB.
- The gate still failed because the default +4 dB dense-electronic item did not qualify repeated level metrics, and elevated settings showed terminal-pressure observations plus RMS retreat on selected material.
- This is not automatic .12 approval. It narrows the next question to whether elevated-setting level retreat is audible as punch loss, bass blur, limiter pumping, low-end crowding, or fatigue.

Validation

- The Node harness completed the targeted map and recorded the gate in local run state.
- Raw WAV captures and generated JSON/Markdown reports remain local-only.
- No EEL DSP scripts changed.

Next Recommended Work

- Interpret the completed .11 follow-up before proposing .12.
- If further action is justified, create a narrow listening target around elevated Sub Harmonics punch/level retreat rather than normal-material clipping.
- Keep PR publication and merge gated on explicit approval.

Run 024 - ‘.11‘ Sub Harmonics Interpretation

Date: 2026-06-08
Status: Completed
Scope: No-candidate-yet interpretation of the completed .11 Sub Harmonics follow-up and focused listening target definition.

What Was Implemented

- Added docs/sub-harmonics-interpretation-v4.1.4.11.md.
- Recorded the decision to keep v4.1.4.11 accepted and not create .12 from the measurement alone.
- Defined a focused listening target for accepted .11 at +4, +10, and +12 dB Sub Harmonics settings.
- Updated docs/system-status.md, docs/task-backlog.md, and docs/README.md so the interpretation record is discoverable.

Why It Matters

- The completed real-JDSP follow-up now has an explicit engineering decision: it found no normal-material clipping through +12 dB, but it did find elevated-setting RMS retreat and terminal-pressure observations that should be checked by ear before candidate work.
- A future .12 must be justified by a repeatable audible defect such as kick softening, bass blur, limiter pumping, low-end crowding, fatigue, or practical loudness loss.
- The next step is structured listening on accepted .11, not EEL editing.

Validation

- Documentation-only change.
- No EEL DSP scripts changed.
- No .12 candidate was created.

Next Recommended Work

- Run focused accepted-.11 listening at +4, +10, and +12 dB.
- Use the structured listening-record fields added in Run 022.
- Keep publication, merge, and candidate creation gated on explicit approval.

Run 025 - ‘.11‘ Sub Harmonics Listening Target Template

Date: 2026-06-08
Status: Completed
Scope: Focused accepted-.11 listening target and local-copy listening-record template for the Sub Harmonics follow-up.

What Was Implemented

- Added docs/sub-harmonics-listening-target-v4.1.4.11.md.
- Added docs/templates/sub-harmonics-listening-record-v4.1.4.11.json.
- Linked both files from docs/README.md.
- Updated docs/system-status.md and docs/task-backlog.md so the next step is the focused accepted-.11 listening check.

Why It Matters

- The .11 follow-up now has a concrete listening workflow instead of only a measurement interpretation.
- The JSON template is designed to be copied into local state, filled with private listening notes, and validated with scripts/validate_axiom_listening_record.py.
- The repo remains safe: it contains prompts, material classes, and public workflow instructions, not private song titles or local listening results.

Validation

- Documentation/template-only change.
- No EEL DSP scripts changed.
- No .12 candidate was created.

Next Recommended Work

- Copy the listening-record template into local state and perform the accepted .11 +4, +10, and +12 dB listening pass.
- If no repeatable audible defect is heard, close the .11 investigation without .12.
- If a repeatable defect is heard, draft a narrow .12 hypothesis before any EEL work.

Run 026 - Confirmatory ‘.11‘ Sub Harmonics Map

Date: 2026-06-09
Status: Completed
Scope: Confirmatory real-JDSP map of accepted .11 at +4, +10, and +12 dB Sub Harmonics using the existing investigation and local material filter.

What Was Implemented

- Ran the documented map-sub-gain command for 20260603T004349-post-acceptance-v4-1-4-1-0d309b.
- Used the accepted src/axiom_binaural_dsp_v4.1.4.11.eel file and temporary external slider fixtures only.
- Updated docs/sub-harmonics-follow-up-v4.1.4.11.md with the repo-safe rerun summary.
- Updated docs/sub-harmonics-interpretation-v4.1.4.11.md and docs/system-status.md so the latest result is discoverable.

Why It Matters

- The rerun confirmed that normal material stayed unclipped through +12 dB.
- The map still failed, but the hard failure was the default +4 dB dense-electronic repeatability qualification, not normal-material clipping.
- Elevated +10 dB and +12 dB again showed repeatable RMS retreat on hip-hop/trap-sub and bass-light material.
- The engineering decision remains unchanged: keep .11 accepted, do not create .12 yet, and use focused listening to decide whether the measured retreat is audible as a real defect.

Validation

- Real-JDSP map completed and wrote a full local aggregate report.
- Documentation-only repo change.
- No EEL DSP scripts changed.
- No .12 candidate was created.

Next Recommended Work

- Perform the accepted .11 focused listening pass at +4, +10, and +12 dB.
- Fill and validate the local listening record.
- Create a .12 hypothesis only if listening finds a repeatable normal-material defect.

Run 027 - Filtered Sub Harmonics A/B Listening Packages

Date: 2026-06-09
Status: Completed
Scope: Tooling and local package workflow for accepted .11 Sub Harmonics listening at +4 versus +10 and +4 versus +12.

What Was Implemented

- Added --include-regex and --exclude-regex filters to scripts/build_axiom_ab_listening_package.py.
- Added API and CLI test coverage for filtered package selection.
- Updated docs/ab-listening-packages.md with the filter workflow.
- Updated docs/tool-inventory.md to mention the A/B package filters.
- Updated docs/sub-harmonics-listening-target-v4.1.4.11.md with optional local package commands for the Sub Harmonics listening pass.
- Updated docs/task-backlog.md so AX-TASK-022 reflects the prepared local A/B package workflow.
- Built local filtered packages from the completed .11 Sub Harmonics map: +4 versus +10 and +4 versus +12, excluding flawed stress material.

Why It Matters

- The listening step can now use blinded A/B files with gain recommendations instead of relying only on manual slider switching.
- Intentional flawed-source stress material can be excluded from normal-material acceptance listening without copying or pruning capture trees by hand.
- The local packages directly target the measured concern: whether elevated Sub Harmonics settings sound quieter, softer, blurrier, or more fatiguing after level handling.

Validation

- python3 -m unittest tests.test_build_axiom_ab_listening_package
- python3 -m unittest discover -s tests -p 'test_*.py'
- Generated +4 versus +10 package: PASS_WITH_WARNINGS
- Generated +4 versus +12 package: PASS_WITH_WARNINGS
- No EEL DSP scripts changed.
- No .12 candidate was created.

Next Recommended Work

- Open the local listening package folder and audition the blinded A/B pairs.
- Fill and validate the local listening record.
- Treat any result as listening evidence for the .11 investigation, not as automatic .12 approval.

Run 028 - Consolidated Local Review Helper

Date: 2026-06-09
Status: Completed
Scope: Initial implementation of AX-TASK-026, a safe non-JDSP local review snapshot for Axiom Codex sessions.

What Was Implemented

- Added local-review to tools/axiom-codex/axiom_codex.py.
- Registered local-review in tools/axiom-codex/command_surface.json.
- Added helper tests for command registration, JSON output, Markdown output, report-file writing, and non-JDSP boundaries.
- Updated docs/task-backlog.md to mark AX-TASK-026 as initially complete.
- Updated docs/tool-inventory.md and docs/codex-operating-guide.md so the command is discoverable.

Why It Matters

- A fresh Codex session can now run one command to collect git state, accepted baseline identity, changed paths, guard-check, ready-check, Knowledge audit, Python tests, Node harness tests, and a recommended next action.
- The command is intentionally non-JDSP, so it can be used for orientation and review without touching the audio route.
- JSON and Markdown report output make it suitable for future dashboards, handoffs, or PR summaries.

Validation

- python3 -m unittest tests.test_axiom_codex_helper
- Command-surface JSON validation.
- python3 tools/axiom-codex/axiom_codex.py local-review
- python3 tools/axiom-codex/axiom_codex.py local-review --skip-tests --skip-knowledge

Next Recommended Work

- Use local-review as the standard orientation command before larger Axiom development batches.
- Consider a machine-readable task-state file next, so local-review and a future next-action command can reason beyond Markdown tables.

Run 029 - Machine-Readable Task State And Next Action

Date: 2026-06-09
Status: Completed
Scope: Initial implementation of AX-TASK-027 and AX-TASK-028, a structured task-state layer and command-backed next-action helper for Axiom Codex planning.

What Was Implemented

- Added tools/axiom-codex/task_state.json as the machine-readable task index.
- Added task-state to tools/axiom-codex/axiom_codex.py.
- Added next-action to tools/axiom-codex/axiom_codex.py.
- Registered task-state in tools/axiom-codex/command_surface.json.
- Registered next-action in tools/axiom-codex/command_surface.json.
- Added task-state validation to local-review.
- Added next-action planning output to local-review.
- Added helper tests for task-state validation, next-action output, and CLI behavior.
- Updated docs/task-backlog.md, docs/tool-inventory.md, and docs/codex-operating-guide.md.

Why It Matters

- Agents can now inspect task phase, blocked state, approval requirements, and recommended next actions without parsing only Markdown tables.
- next-action can now recommend safe work while respecting dirty working trees, listening blockers, and explicit approval gates.
- The task metadata keeps .11 Sub Harmonics correctly marked as blocked on human listening while keeping Knowledge concept-note work available as a safe Agentic Layer next step.

Validation

- python3 -m unittest tests.test_axiom_codex_helper
- python3 tools/axiom-codex/axiom_codex.py task-state
- python3 tools/axiom-codex/axiom_codex.py next-action --json

Next Recommended Work

- Use local-review, task-state, and next-action as the standard Agentic Layer orientation sequence.

Run 030 - Knowledge Concept Notes

Date: 2026-06-10
Status: Completed
Scope: Complete AX-TASK-024 by converting seed Knowledge sources into repo-safe Axiom concept notes for test design.

What Was Implemented

- Added docs/knowledge/concepts/spatial-listening-vocabulary.md.
- Added docs/knowledge/concepts/elevated-bass-headroom-tradeoff.md.
- Added docs/knowledge/concepts/stage-isolation-and-fixture-scope.md.
- Added docs/knowledge/concepts/reproduction-boundaries-and-profile-scope.md.
- Linked the concept notes from docs/knowledge/README.md.
- Updated docs/task-backlog.md and tools/axiom-codex/task_state.json so AX-TASK-024 is complete.

Why It Matters

- Axiom Knowledge now has a concept layer between broad bibliography notes and concrete engineering work.
- Future agents can use the notes as controlled vocabulary for listening records, fixture design, Sub Harmonics interpretation, and profile-boundary decisions.
- The notes remain repo-safe: they cite Source IDs and summarize Axiom use in original wording without copying protected source text or private paths.

Validation

- python3 tools/axiom-codex/axiom_codex.py knowledge-sources
- python3 tools/axiom-codex/axiom_codex.py knowledge-query "Sub Harmonics bass punch listening"

Next Recommended Work

- Run next-action after validation; with AX-TASK-024 complete, the remaining major open items are blocked on human listening or explicit approval.

Run 031 - PR 12 Merge

Date: 2026-06-10
Status: Completed
Scope: Approved publication/merge of the Codex hardening, Knowledge, Sub Harmonics follow-up, local-review, task-state, and next-action batch.

What Was Implemented

- Ran local-review before publication; it passed.
- Pushed codex/axiom-codex-hardening-knowledge to GitHub.
- Marked PR #12 ready for review.
- Merged PR #12 into main after explicit user approval.
- Updated docs/task-backlog.md and tools/axiom-codex/task_state.json so AX-TASK-025 is complete.

Why It Matters

- The Agentic Layer work is now on main.
- main now includes the Axiom Codex helper command surface, role profiles, guard checks, skill evals, local review, task-state, next-action planning, Knowledge seed notes, Knowledge concept notes, and the .11 Sub Harmonics follow-up documentation.
- The remaining major Axiom task is the .11 Sub Harmonics listening decision, which still requires human listening rather than automated approval.

Validation

- python3 tools/axiom-codex/axiom_codex.py local-review
- PR #12 was mergeable and had no GitHub check failures configured.

Next Recommended Work

- Close AX-TASK-022 only after actual listening evidence exists.

Run 032 - Agentic Contract Hardening

Date: 2026-06-21
Status: Completed
Scope: First hardening batch for AX-TASK-018 through AX-TASK-021.

What Was Implemented

- Added the agentic-audit helper command.
- Added strict command-surface validation for required fields, duplicate names and aliases, helper-to-runtime mappings, and JDSP approval boundaries.
- Added profile validation for role-source mappings and required sections.
- Added skill-eval schema, duplicate-ID, command-mapping, and term validation.
- Integrated these contracts into ready-check and local-review.
- Added adversarial regression tests and synchronized the installed $axiom-engineering skill.
- Added AX-TASK-034 as the completed contract-audit milestone.

Why It Matters

- Agentic registries are now validated runtime inputs rather than documentation-only files.
- Unsafe approval metadata and command/runtime drift fail before publication.
- Future role and behavior-fixture changes have deterministic structural gates.

Validation

- python3 -m unittest discover -s tests -p 'test_*.py' passed 206 tests.
- npm test under tools/axiom-team passed 23 tests.
- python3 tools/axiom-codex/axiom_codex.py agentic-audit passed.
- python3 tools/axiom-codex/axiom_codex.py local-review --no-untracked passed.
- python3 tools/axiom-codex/axiom_codex.py guard-check --json returned no findings.

Next Recommended Work

- Replace the current free-form agent-review scaffold with a validated, machine-readable multi-role findings record.

Run 033 - Validated Agent Review Records

Date: 2026-06-21
Status: Completed
Scope: Replace the free-form agent-review scaffold with a validated machine-readable multi-role review record.

What Was Implemented

- Added an axiom-agent-review schema for draft multi-role reviews.
- Added agent-review --json and agent-review --output <record.json>.
- Linked each review role to its Codex profile and Pi role source.
- Added decision enums, forbidden-scope boundaries, evidence-status markers, findings arrays, and evidence-needed arrays.
- Added validation for unknown roles, duplicate roles, malformed decisions, and missing record fields.
- Added AX-TASK-035 as the completed multi-role review-record milestone.

Why It Matters

- Future agents can produce structured review artifacts that are both readable and machine-checkable.
- Draft review records are explicitly marked as not evidence until findings are completed.
- Multi-role review output can now feed future orchestration without parsing a loose Markdown scaffold.

Validation

- python3 -m unittest tests.test_axiom_codex_helper
- python3 tools/axiom-codex/axiom_codex.py agent-review --topic "Agentic Layer review contract" --json
- python3 tools/axiom-codex/axiom_codex.py agentic-audit
- python3 tools/axiom-codex/axiom_codex.py ready-check

Next Recommended Work

- Improve next-action so it can intentionally select from initial-maintenance Agentic work instead of reporting that no unblocked task is available.

Run 034 - Maintenance-Aware Next Action

Date: 2026-06-21
Status: Completed
Scope: Harden next-action so Agentic maintenance work can be selected explicitly without weakening default planning boundaries.

What Was Implemented

- Added next-action --include-maintenance.
- Kept initial-maintenance tasks excluded from default next-action.
- Preserved dirty-tree, evidence-failure, approval-gate, blocker, and seeded task protections.
- Added machine-readable includeMaintenance output.
- Updated the Markdown output to render the full boundary list.
- Added AX-TASK-036 as the completed maintenance-aware planning milestone.

Why It Matters

- Agents can now continue Agentic hardening without manual interpretation of the open task list.
- Normal planning remains conservative and will not treat initial-maintenance work as generally actionable.
- The selected maintenance task is visible even when the current working tree must be validated first.

Validation

- python3 -m unittest tests.test_axiom_codex_helper
- python3 tools/axiom-codex/axiom_codex.py next-action --include-maintenance --no-evidence
- python3 tools/axiom-codex/axiom_codex.py agentic-audit

Next Recommended Work

- Use next-action --include-maintenance --no-evidence after this commit to pick the next Agentic hardening target.